14 The reaction between substances **A** and **B** is found to follow the rate law

$$\text{rate} = k[\mathbf{A}]^{m}[\mathbf{B}]$$

where k is the rate constant and has units of $mol^{-2}\ dm^{6}\ s^{-1}$.

Two experiments to study the kinetics of this reaction were carried out and the data obtained are tabulated below.

Experiment	Initial [**A**] / $mol\ dm^{-3}$	Initial [**B**] / $mol\ dm^{-3}$	Initial rate / $mol\ dm^{-3}\ s^{-1}$
1	0.040	0.080	R
2	0.020	y	$\frac{R}{2}$

What is the value of y in experiment 2?

A 0.020 **B** 0.040 **C** 0.160 **D** 0.320